

AUTOMATED RESUME SCREENING SYSTEM USING NLP AND COSINE SIMILARITY

Authors: BONDOC, PATRICIA A. | REYES, CARL J.

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Department of Information Systems

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ABSTRACT

This study builds an automated resume screening system using TF-IDF and cosine similarity for job-resume matching, achieving 84.7% matching accuracy and reducing HR screening time by 73% in a company pilot.

INTRODUCTION

HR departments in Philippine companies face high applicant volumes, making manual resume screening time-intensive and inconsistent. Automated screening tools can rank candidates by relevance to job descriptions. This study develops a web-based screening system for a BPO company in Dasmariñas City with 200 average monthly applicants.

METHODOLOGY

TF-IDF vectorization was applied to both job descriptions (JD) and resumes after preprocessing: stopword removal, stemming (NLTK), and skills entity extraction using spaCy's NER model fine-tuned on a resume corpus. Cosine similarity computed between JD and resume vectors produces a match score. A threshold of 0.45 was used to classify candidates as Shortlisted or Not Shortlisted. Pilot: 3 job openings, 120 applicants, evaluated against HR manager decisions.

RESULTS

The system achieved 84.7% accuracy against HR manager ground-truth decisions, precision 82.1%, recall 87.3%. Screening time was reduced from 3.5 minutes to 0.6 minutes per resume (73% reduction). Top candidates identified by the system produced 78% offer acceptance rate vs. 65% historical baseline. Skills NER extraction accuracy was 91.6%.

DISCUSSION

TF-IDF cosine similarity provides a strong, interpretable baseline for job-resume matching. The 84.7% accuracy ceiling is partly explained by subjective HR preferences not fully captured in textual matching (e.g., cultural fit). Future work should extend to BERT-based semantic matching and incorporate structured resume parsing for education and experience extraction.